

Deepak Mallampati

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US-based (Kent, OH), F-1 OPT | Open to up to 25% travel

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping production RAG, agentic LLM workflows, and predictive ML pipelines for healthcare and enterprise contexts — including enterprise Order-to-Cash data engineering at Emerson Synthesis (oil & gas finance, contracts, allocations, invoicing). Comfortable owning end-to-end delivery: data, prompts, models, evaluation, FastAPI/Flask APIs, and Docker-based deployment. Looking to apply this build-and-implement discipline as an AI Forward Deployed Engineer where customer outcomes drive the engineering loop.

CORE COMPETENCIES

AI Forward Deployed Engineering • Enterprise Software Delivery • LLM Application Development • RAG Engineering • Agentic LLM Workflows • Predictive ML Pipelines • Forecasting (walk-forward validation) • Python • SQL (T-SQL, PostgreSQL) • FastAPI • Flask • REST APIs • Docker • CI/CD (Jenkins) • Data Modeling • ETL/ELT • Data Visualization (Grafana, Excel) • Customer-facing Stakeholder Docs • Data Privacy & Governance (HIPAA-conscious, RBAC, audit logging)

PROFESSIONAL EXPERIENCE

Jr. AI/ML Engineer Trainee — Progress Solutions Inc.

Jul 2025 – Present | USA | Healthcare AI / SaaS

- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved contextual retrieval precision **~35%** and reduced irrelevant retrieval **>30%**.
- Developed **agentic LLM workflows** with structured reasoning, tool discipline, and grounding rules; improved agent response stability **~25%**; designed reusable agent patterns and failure-recovery contracts that lift across new healthcare verticals.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement, and support prioritization; recall **+15–20%** on high-risk categories at precision **>90%**.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**; reduced post-deployment defects **~30%**.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; raised dataset reliability **>98%**.
- Authored **stakeholder-facing system-limitation docs** alongside API rollout — the customer-facing artifact that drives adoption and surfaces feedback.
- Maintained HIPAA-conscious data governance: de-identification, lineage documentation, evaluation audit trails.

Database & DevOps Performance Engineer (Intern) — Energy Solutions International (Emerson)

Jun 2022 – Apr 2023 | Hyderabad, India | Enterprise ERP — Order-to-Cash

- Optimized **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform (contracts, nominations, allocations, invoicing — **enterprise financial workflows** directly adjacent to CPM/EPM); cut query execution time **~20%** and data retrieval latency **~25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates, version-controlled SQL scripts, and stored procedure deployments; reduced deployment errors **>30%** and improved release cycle time **~35–40%**.
- Designed performance dashboards using **SQL Server DMVs and Grafana** to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence **~25%**.
- Tuned database maintenance (index rebuilds, statistics updates, partition-aware indexing); improved reconciliation job runtime **~15%**.
- Supported **RBAC implementation and audit logging** for financial modules in a compliance-sensitive oil & gas environment.

Machine Learning Engineer — Suvidha Foundation

Jan 2022 – Mar 2022 | Hyderabad, India

- Built a **transformer-based video summarization** system across 5,000+ recorded sessions; reduced manual review time **60–70%**.
- Implemented **RAG document Q&A**; reduced hallucinated outputs **~30%**; deployed as **Flask API**.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Risk) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. Reusable JSON-contract pattern — same design-for-reuse instinct enterprise AI Forward Deployed roles need for customer-customizable blocks (Genesis Blocks / Marketplace).

Manga Lens — Multi-Provider AI Vision Chrome Extension (*shipped, Chrome Web Store*)

Solo-built extension integrating **four AI vision providers** with provider abstraction layer, per-provider payload routing, 7-day cache, 29 site selector configs. Customer feedback loop via Chrome Web Store reviews shaped multi-provider fallback design.

Dream Decoder — Multimodal Creative Intelligence Platform

FastAPI + React/TypeScript app with intermediate **structured prompt transformation layers**; improved contextual alignment ~**30%** and first-pass image success ~**25–30%**.

IoT + ML Smart Building Indoor Temperature Prediction

Time-series forecasting with walk-forward validation comparing Random Forest, SVR, Linear Regression; lag-window feature engineering; Random Forest achieved ~**20–30%** lower MAE vs linear baseline.

EDUCATION

Master's Degree — Kent State University (*Aug 2023 – May 2025*) | *Ohio, USA*

CERTIFICATIONS

Available upon request

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP — *willing to ramp C# during implementation projects*

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex

Data: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, ETL/ELT, data modeling

Infra & DevOps: Docker, Jenkins (CI/CD), GitHub, Grafana, RESTful APIs, structured logging, MLOps/LLMOps, cloud-ready deployment (Azure-adjacent)

Domains: Enterprise data platforms (Order-to-Cash, financial workflows), CPM/EPM-adjacent SQL + ETL, Healthcare AI (HIPAA-conscious), Workflow Automation